Alejandro Bay

Full Stack Software Engineer
| Python | AI | Typescript | React | MS Azure |Java | C# | Brooklyn, NY
LinkedIn | GitHub | Portfolio | Twitch alejandrobay23@gmail.com

I am a Full Stack Software Engineer specializing in **Python, FastAPI, AI** implementation, **TypeScript**, **Node, React, Redux, Unity (C#) SQL, Pytest** and **Java** with **Spring**. I have a strong background in **backend systems**, **REST APIs**, and real-time dashboards. With 7 years of experience as a Business Analyst**,** I bring a data-driven mindset to development, emphasizing KPIs and automation. I have built and deployed full-stack web apps using the **MERN stack**, **Java**, and **Python**. Current interests in learning more about **LLMs**, **AI** chatbot implementations, **WebSocket** and **Microservices** with **Python** and **Java**.

Experience

Freelance Full Stack Web Developer
New York, NY | Jan 2022 – Present
Built and deployed full-stack web apps using **React**, **Next.js**, **Node.js**, **FastAPI**, and **Java**.

- Integrated **REST APIs** with real-time dashboards and **TypeScript** form validation.
- Designed scalable components using modular architecture and **TailwindCSS**.
- Debugged production codebases and implemented secure authentication workflows.

Operations Manager
Wildlife Conservation Society (WCS), New York, NY | Nov 2018 – Present
Led operations at the NY Aquarium generating $1.5M+ annual revenue with a team of 45+ employees.

- Created internal MERN stack app for inventory and KPI tracking.
- Increased revenue by 12% YoY and delivered 32% operating income.
- Built forecasting tools and dashboards using **Python**, **Excel**, **MS Automate**, **MS SharePoint** and **Power BI**.

Business Support Coordinator
Sanofi, Caracas, Venezuela | 2010 – 2015
Coordinated marketing technology initiatives and KPI systems across departments.

- Spearheaded the launch of an **iOS AR app**, bridging dev and business teams.
- Delivered dashboards that improved campaign efficiency **IBM Cognos**.

Senior Business Analyst
Atento (Telefonica), Caracas, Venezuela | 2007 – 2010
Owned analytics for sales and operations across Telefonica accounts.

- Identified $1M+ in upsell opportunities through data insights.
- Redesigned CRM workflows, reducing sales cycle times by 20%.

Projects

Papelon Inventory Management System
React, Node.js, Express, MongoDB, Redux, TailwindCSS
GitHub | Live Demo

- Built an inventory control app with **role-based access**, dynamic SKUs, and KPI dashboards.
- Created **secure REST APIs**, implemented **JWT auth**, and reduced reporting effort by 80%.

Lego Work Orders Tracker
React, Node.js, Express, MongoDB, TailwindCSS, Redux, TypeScript.
GitHub | Live Demo

- Designed internal app for service request management at NY Aquarium.
- Added **status filtering**, lifecycle tracking, and secure login.

AI Chatbot Assistant
Python, FastAPI, OpenAI API, LangChain, WebSocket
GitHub

- Developed AI chatbot using **FastAPI**, **LangChain**, and **OpenAI GPT APIs**.
- Integrated document loading, and context memory.

Education & Certifications

Languages
Fluent in English and Spanish.

Bachelor of Science in Computer Science (Thesis Completed)
New Professions Technical Institute, Caracas, Venezuela — 2014

Certifications

- Enterprise Design Thinking Practitioner (IBM) – 2024
- Generative AI Fundamentals (Google) – 2024
- Learning Spring Boot 3 with Spring 6 (LinkedIn Learning) – 2024
- Project Management Certificate (Cornell) – 2022

Interests

AI & LLMs | Game Development with **Unity (C#)** | **Microservices** | Serverless Architecture | Data Visualization | SaaS Applications | Live Streaming